CURRICULUM MODULE (CORE ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 • March 2026
Confidential • Internal Documentation

Audience: School Administrators • Teachers • IT Team • Students • Parents
Related Modules: Academics • Student Management • Assessments • Learning Resources • Parent Portal • Analytics

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1. MODULE OVERVIEW

1.1 Purpose

The Curriculum Module provides a structured framework for planning, monitoring, and managing all academic content delivery across subjects and lessons. It allows schools to:

- Track curriculum coverage by grade, subject, and week
- Align Schemes of Work (SOW) with lesson plans
- Integrate teaching resources and learning materials
- Link topics to assessments for student performance tracking
- Support teacher collaboration and planning
- Provide actionable insights for administrators and parents

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2. CORE OBJECTIVES

1. Enable teachers to plan and upload Schemes of Work
2. Monitor weekly lesson coverage and pacing
3. Integrate learning resources directly into curriculum topics
4. Link topics to assessments and track student mastery
5. Facilitate teacher collaboration on lesson planning and resource sharing
6. Provide parent and student visibility of curriculum progress
7. Support offline access and mobile-first interaction
8. Offer analytics for curriculum compliance, weak topics, and resource gaps

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3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

Curriculum Created → Subjects Defined → Topics Assigned → Schemes of Work Uploaded → Resources Linked → Lessons Delivered → Assessments Conducted → Performance Tracked → Progress Reports Generated

3.2 Daily Operational Flow

Teacher reviews weekly plan
↓
Lessons delivered as per Schemes of Work
↓
Resources accessed or assigned to students
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Assessments linked to topics are administered
↓
System calculates topic coverage and student mastery
↓
Alerts generated for overdue topics or lagging subjects

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4. MODULE ARCHITECTURE

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Curriculum Module
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├── Curriculum Mapping (Subjects, Lessons, Topics)
├── Schemes of Work Management
├── Lesson Plan Integration
├── Learning Resource Library
├── Assessment Linkage
├── Teacher Collaboration Tools
├── Parent & Student Portal Access
├── Analytics & Reporting
├── Alerts & Notifications
├── Offline & Mobile Access
├── Security & Role-Based Access
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5. CURRICULUM MAPPING

5.1 Features

- Define subjects and topics per grade level
- Assign lesson sequences and pacing per week
- Monitor coverage percentage per topic and subject
- Quick overview of subjects on track vs lagging

5.2 Data Model

Curriculum

- curriculum_id
- subject_id
- grade_level
- topic_id
- week_number
- coverage_percentage

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6. SCHEMES OF WORK MANAGEMENT

6.1 Features

- Upload and create Schemes of Work (SOW) per subject
- Link SOW to lessons and topics
- Version control for updates
- Track weekly progress against planned coverage

6.2 Data Model

SchemeOfWork

- sow_id
- subject_id
- grade_level
- topics_covered
- week_number
- version
- status (planned/delivered)

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7. LEARNING RESOURCE LIBRARY

7.1 Features

- Upload PDFs, videos, e-books, and interactive content
- Link resources to specific topics or lessons
- Searchable and filterable by subject or grade
- Teacher and admin permissions for access and editing

7.2 Data Model

Resource

- resource_id
- title
- type (PDF, video, interactive, etc.)
- linked_topic_id
- uploaded_by
- access_level

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8. ASSESSMENT LINKAGE

8.1 Features

- Assign quizzes, tests, or exams to topics
- Track student performance per topic
- Flag weak areas for intervention
- Integrate with Academics Module for grade calculation

8.2 Data Model

AssessmentLink

- assessment_id
- topic_id
- student_id
- score
- max_score
- term
- comments

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9. TEACHER COLLABORATION

9.1 Features

- Shared lesson plan editing
- Peer review and commenting
- Resource sharing across teachers and subjects

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10. PARENT & STUDENT PORTAL

10.1 Features

- View curriculum progress
- Access linked resources
- Track completed topics vs planned
- Notifications for missing lessons or upcoming assessments

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11. ANALYTICS & REPORTING

11.1 Features

- Track subject and topic coverage percentages
- Identify weak subjects and overdue topics
- Monitor resource usage gaps
- Provide historical trend reports for administrators

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12. ALERTS & NOTIFICATIONS

12.1 Triggers

- Topics falling behind schedule
- Missing Schemes of Work or resources
- Upcoming assessments
- Low coverage or student performance alerts

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13. SECURITY & ROLE-BASED ACCESS

Role	Access Scope
Admin	Full access, curriculum setup, reports
Teacher	Assigned subjects and topics, resources, assessments
Parent	Own child progress overview
Student	View assigned lessons and linked resources

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14. MOBILE-FIRST & OFFLINE DESIGN

14.1 Requirements

- Mobile-friendly lesson and topic navigation
- Offline mode with downloadable lesson packs and resources
- Auto-sync when connectivity is restored

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15. SCALABILITY

- Handle curriculum tracking for multiple campuses and thousands of students
- Quick data retrieval and analytics even for large topic hierarchies
- Extendable to new curricula, topics, and learning resources